

Gray2ColorNet: Transfer More Colors from Reference Image

Peng Lu¹, Jinbei Yu¹, Xujun Peng², Zhaoran Zhao¹, Xiaojie Wang¹

¹School of Computer Science, Beijing University of Posts and Telecommunications, Beijing, China

²ISI, University of Southern California, Marina del Rey, CA, USA

lupeng@bupt.edu.cn, jimbo@bupt.edu.cn, xpeng@isi.edu, zhaozhaoran@bupt.edu.cn, xjwang@bupt.edu.cn

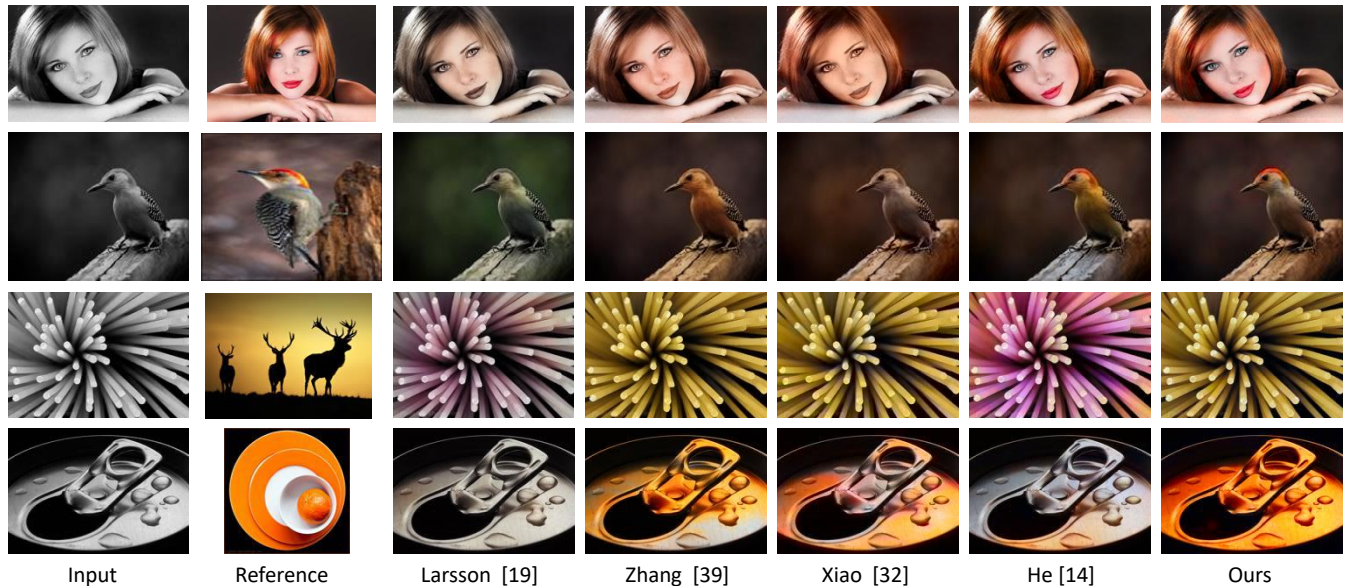


Figure 1: Colorization results using the state-of-the-art approaches. The proposed method generates plausible color images with input grayscale images which are related or independent to the reference image semantically.

ABSTRACT

Image colorization is an effective approach to provide plausible colors for grayscale images, which can achieve better and pleasing visual qualities. Although exemplar based colorization approaches provide promising results, they are relied on semantic colors or global colors only from the reference images. For the former situation, when the correspondence between the input grayscale image and reference image is not established, the colors of the reference image cannot be transferred to the input grayscale image successfully. With the later circumstance, because only global colors are considered, it is hard to produce a color image whose objects have the same color as the reference image when they are semantically related. Thus, an end-to-end colorization network (**Gray2ColorNet**) is proposed in this work, where an attention gating mechanism

based color fusion network is designed to accomplish the colorization tasks. Relied on the proposed method, the semantic colors and global color distribution from the reference image are fused effectively, which are transferred to the final color images along with the prior knowledge of colors contained in the training data. The experimental results demonstrate the superior colorization performances of the proposed method compared to other state-of-the-art approaches.

CCS CONCEPTS

• Computing methodologies → Computational photography.

KEYWORDS

Image understanding, GAN, Colorization

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1 INTRODUCTION

Image colorization, which adds colors to grayscale images to improve their visual quality and make them more pleasing, plays an important role for a wide range of image processing applications.

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However, as a well known ill-posed problem, where multiple different colors are potentially acceptable for a single pixel within the input image, colorization is still a challenging task. Thus, in order to overcome this ambiguity difficulty, colorization with human intervention attracts increasing research interests recently.

To fulfill the image colorization based on user's interaction, user-guided scribbles and sample reference images are two widely used clues.

The user-guided colorization employs color strokes provided by users on particular objects in the image, where those colors are propagated across the entire image relied on simple similarity metrics between neighboring pixels or regions with optimization approaches [16, 21, 23, 27]. In recent years, several deep neural network based methods are proposed, which combine the prior colors learned from the large scale data and the user interactions to achieve colorization [29, 39]. As one can see, these approaches demand a certain amount of human efforts to label the key objects' colors in the image, which requires deep understanding of colors from the user and limits its effectiveness.

For the reference image (or exemplar) based colorization, a reference image whose tones and colors are favored by the user is provided, from which the colors are transferred to the input grayscale images. Because exemplar based approaches can effectively represent user's preference and requirements with less annotation efforts, it attracts more research interests recently and we prefer to focus our colorization approach on this trend.

Among many existing exemplar based colorization methods, some of them tend to use global tones of the reference image to accomplish the colorization tasks [32]. However, if only global tones of the reference image are employed, the objects in the input image might be mis-colored, which can be seen from sample outputs of Zhang *et al.*'s approach [39] and Xiao *et al.*'s method [32] in the first row of Fig. 1. Thus, more researches prefer to apply the semantic correspondence between input image and reference image initially. Then the colors are transferred from reference image to the input image afterwards [2, 3, 6, 12, 13, 17, 35]. But the failure of finding correspondence or lack of similar objects between input and reference images are the main obstacles for the success of this type of exemplar based colorization. Although the state-of-the-art method [14] uses the prior color knowledge of the large scale image dataset to alleviate this problem, where the missing correspondence between input image and reference image can be found in the training dataset, it still has several shortcomings. Firstly, two related tasks semantic correspondence calculation and image colorization are learned in two network independently, which degrades the system performance and makes the overall system more complex. Secondly, it ignores the user's preference and degrades the pleasing experience from the users when the semantic correspondence is lacking between input grayscale image and reference image. This phenomenon can be easily observed from the sample output of method [14] in the third row of Fig. 1.

From Fig. 1 we can see the semantic colors associated with the objects and global color distribution such as tones are both important color characteristics of the reference image. For a reliable exemplar based colorization system, these two types of color information should be considered simultaneously. However, extracting and combining these different types of colors in a unified network

for colorization is a challenging task due to the ambiguous nature of them, where each single pixel's color may be affected by the color from corresponding area of the reference image or the overall tone of the reference image.

In this paper, an end-to-end deep neural network named as Gray2ColorNet is first proposed which fuses the semantic colors and global colors of reference image along with the prior knowledge of colors contained in the large-scale dataset for effective colorization. The proposed Gray2ColorNet consists of two parts: gated color fusion sub-network (GCFN) and main colorization sub-network (MCN). The former can adaptively fuse the semantic color information and color distribution information provided by the reference image through the attention based gating mechanism. The latter combines the prior knowledge of colors contained in the large-scale training data and the color features of the reference image produced by GCFN to achieve the final color image. Based on the proposed network, more effective color transfer can be obtained which can also be seen in the last column of Fig. 1.

In summary, we make four contributions to the literature: (1) To the best of our knowledge, the proposed Gray2ColorNet is the first deep neural network that fuses the semantic colors and global colors of reference image for colorization, which effectively transfers the colors from reference image to the input grayscale image; (2) By using multi-task and attention operation on the intermediate layers of the encoder in the main colorization sub-network, the semantic correspondence calculation and image colorization can be learned jointly to boost system's performance, where the extra semantic feature extractor is not needed; (3) A gating mechanism is proposed to automatically determine the importance of semantic colors and global colors and reduces the ambiguity between them; (4) A differentiable histogram generation algorithm is proposed, where the color histogram loss can be applied on the *ab* channel generated by the proposed system directly for effective colorization.

2 RELATED WORK

Without considering hand-crafted rule based image colorization techniques, the modern colorization algorithms can be roughly grouped into three main categories: user-guided, exemplar based, and automatic colorization.

Fully automatic methods performed colorization on grayscale images without requiring user's involvement and realized with neural networks basically. A joint bilateral filtering based post-processing step was proposed by Cheng *et al.* [5] to ensure artifact-free quality. In [10], an object function conditioned on image features was trained to smooth and control the colors in the long spatial range. By considering colorization as an ambiguous problem, a variational autoencoder (VAE) was employed in [9] to produce diverse colorization for a given grayscale image. By feeding grayscale images into the VGG [30], Larsson *et al.* packed the descriptors for each pixel to predict its histogram of hue and chroma, which were utilized to accomplish the final colorization [19]. A self-regularized losses was proposed in [20] to perform video colorization with self-regularization and diversity. These methods, while reducing the burden on users, are not appropriate when considering specific user requirements.

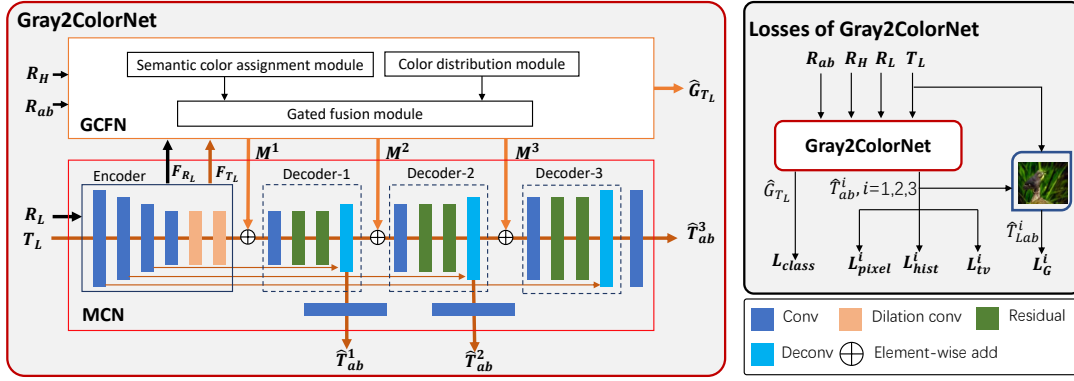


Figure 2: An illustration of the proposed image colorization framework and losses.

To guide the image colorization with user’s interaction, user-guided scribbles and sample reference images are two widely used clues. Traditional user-guided colorization methods normally require users to specify abundant scribbles for the grayscale image to achieve an acceptable result [21], which limits their usage due to intensive user interactions. In [34], a blending approach were designed to allow users obtain desired results based on a small set of chrominance scribbles. By using adaptive clustering in high dimensional affinity space, Xu et al. proposed an optimization approach to propagate the users’ input color hints to the whole image [33]. Similarly, by modeling each pixel as a linear combination of its neighbors in a manifold, the colorization can be accomplished with higher efficiency [4]. More recently, researchers preferred to use deep learning approach for user-guided image colorization, which combined low level image features and high level semantic clues to propagate user hints for grayscale images [7, 29, 39].

Compare to the user-guided, exemplar based colorization only requires the user to provide a single reference image, whose colors are translated to the target grayscale image. Some early works attempted to use matching pixels of specific areas found by the user between a pair of reference and grayscale images, and the colors were transferred for these “seed” regions first then blended to the remaining areas [31]. In [22], Liu et al. used multiple reference images collected from the internet to recover the intrinsic reflectance image. This idea was also employed by Chia et al. but rich texture related features were used for similarity measure and the graph-based optimization approach was applied for color transferring [6]. Unlike exemplar based colorization relied on pixel level matching, Gupta et al. extracted features for superpixels of images to guide the colorization, correspondence confidences between reference and target image were also employed[13]. By heavily leveraging the dense semantic correspondence between images, He et al. proposed a deep neural network matched pixels of two images in high dimensional space and transferred colors from one or more reference images to the target image [14, 15]. And He et. al’s work was extended to video colorization by using recurrent networks [36].

3 PROPOSED METHOD

3.1 Problem formulation

Given a grayscale image $T_L \in \mathbb{R}^{H \times W \times 1}$ which is represented by the luminance channel L and a reference image $R_{Lab} \in \mathbb{R}^{H \times W \times 3}$ which is represented by Lab channels in CIELAB color space, our goal is to find a function $\hat{T}_{ab} = \mathcal{F}(T_L|R_{Lab})$ that maps the input image from grayscale channel to its ab channels of CIELAB color space given reference R_{Lab} , where H, W are height and width of the images.

To accomplish the defined image colorization task, an end-to-end deep neural network named as Gray2ColorNet is designed, which is composed by two sub-networks: gated color fusion sub-network (GCFN) and main colorization sub-network (MCN). The conflicts of semantic colors and global color distribution from the reference image is eliminated by the proposed gated color fusion sub-network. And the fused color features from this sub-network are transferred back to the main colorization sub-network, which are combined with the prior color knowledge eventually to form the final output color image.

Specifically, the MCN extracts the features F_{T_L} and F_{R_L} for input grayscale image T_L and reference image R_L respectively and pass them to GCFN. Based on these features, GCFN uses semantic color assignment module to assign the semantic color features of R_{ab} to the input image initially and then fuses them with the features of color distribution R_H produced by color distribution module through the gated fusion module. In order to increase the semantic representation capability of the features output by the encoder of MCN, which directly affects the accuracy of semantic correspondence calculation and color assignment, a classification task is added to the Gray2ColorNet to predict the semantic label $\hat{G}_{T_L}$ of the input image T_L . Afterwards, the MCN takes the fused color feature M from GCFN with the feature of the input grayscale image T_L extracted by the encoder to produce the ab channels $\hat{T}_{ab}$ of the input image T_L through the decoders. The overall pipeline of the proposed network is illustrated in Fig. 2.

3.2 Main colorization sub-network (MCN)

The main colorization sub-network designed in our work is composed by the classical encoder-decoder structure, which is widely

applied in image generation tasks [11, 14, 40]. Particularly, a stacked network structure with three decoders is employed, whose effectiveness has been proved by [37] for image generation by breaking the entire generative tasks into several sub-tasks of progressive goals through multiple images with different resolutions from its branches, as shown in Fig. 2. The generated image from the deepest branch of the MCN is utilized as our final output color image, but the low resolution images produced at the intermediate branches are also employed to encourage the model to learn the relationships between colors and contents with different scales. Furthermore, we utilize the symmetric shortcut connections between encoder and decoder, which is originated from U-Net [28, 39], to recover the spatial information and avoid vanishing gradient problem for the training.

In our implementation, image features are extracted by an encoder which is consist of a set of conventional conv-relu layers and dilated conv-relu layers, whose kernel sizes are 3×3 and dilated factors are 2. During feature extraction, the tensors are down-sampled by factor of 2 after each convolutional operation. Based on the obtained features, a decoder is employed to reconstruct the color images with different resolutions, which can be decomposed into three sub-decoders. In each sub-decoder, a conv-relu layer is followed by two residual blocks and an up-sampling block with upscale of 2, and the output tensors from each sub-decoder are passed to a conv-relu layer which maps them to two-channel tensors $\hat{T}_{ab}$ to represent ab channels of the output images.

3.3 Gated color fusion sub-network (GCFN)

In gated color fusion sub-network (GCFN), three modules are designed where the semantic color assignment module computes the semantic correspondence between input image and reference image prior to extracting features from R_{ab} and assigns the color features of reference image to the input image based on the obtained semantic correspondence. And the global color distribution features of the reference image are learned through the color distribution module. These two types of color characteristics are fused by the proposed gated fusion module before transferring them back to the main colorization sub-network.

3.3.1 Semantic color assignment module. To model the semantic correspondence between the input image and reference image, the features F_{T_L} and F_{R_L} of them extracted from 6 layers of the encoder in the MCN with different scales are used, where each channel of these features are down-sampled or up-sampled to the same spatial size initially and then concatenated to form two features $\hat{F}_{T_L}$ and $\hat{F}_{R_L}$ respectively, whose final sizes are both $\frac{H}{2} \times \frac{W}{2} \times 1984$.

To fulfill the semantic color assignment task, the dense correspondence can be found by calculating the pairwise similarity between the features $\hat{F}_{T_L}$ and $\hat{F}_{R_L}$. Formally, a correlation matrix C between $\hat{F}_{T_L}$ and $\hat{F}_{R_L}$ is obtained whose element $C(i, j, 1)$ is computed by:

$$C(i, j, 1) = \frac{\hat{F}_{T_L}(i) \cdot \hat{F}_{R_L}(j)}{\|\hat{F}_{T_L}(i)\| \|\hat{F}_{R_L}(j)\|}, \quad (1)$$

where i and j run through the space indexes of feature representation $\hat{F}_{T_L}$ and $\hat{F}_{R_L}$.

Furthermore, the feature representations $F_{R_{ab}}$ of R_{ab} are extracted by two conv-relu layers initially. And then the attended

features F_A can be obtained by re-weighting the feature $F_{R_{ab}}$ based on correlation matrix C through attention operator[1], whose elements are calculated by:

$$F_A(i) = \sum_{j=1}^{HW/4} \alpha_{ij} \cdot F_{R_{ab}}(j), \quad (2)$$

where $1 \leq i \leq \frac{HW}{4}$ and α_{ij} is the weight computed based on:

$$\alpha_{ij} = \frac{\exp(C(i, j, 1))}{\sum_{k=1}^{\frac{HW}{4}} \exp(C(i, k, 1))}. \quad (3)$$

Ultimately, the final output of assigned semantic color features F_S with different spatial sizes are gained by convoluting the F_A with a set of conv-relu layers.

In order to improve the semantic representation capability of the features output by the encoder of MCN, which are crucial to semantic correspondence calculation, the feature map of the last layer of the encoder is also used for the image classification task to predict input image's class label $\hat{G}_{T_L}$.

It should be noted that as the key steps to the semantic based color assignment, the semantic correspondence calculation between input and reference image in this work is implemented in a different way as the state-of-the-art methods [14, 36], which uses the extra feature extractors and makes the whole network structure more complex. In our implementation, the semantic correspondence can be easily obtained by attention operator [1] based on the features extracted from encoder of MCN, where no extra feature extractors is needed. In this way the semantic correspondence calculation and image colorization can learn together and promote each other to further improve assignment accuracy.

The overall structure of the proposed semantic color assignment module is demonstrated in Fig. 3.

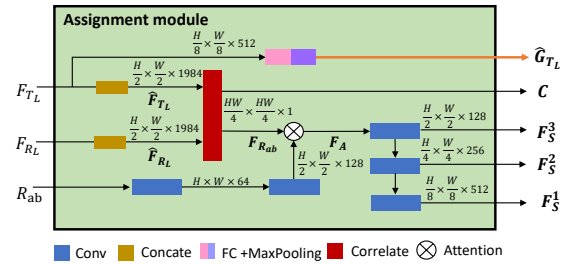


Figure 3: Structure of semantic color assignment module.

3.3.2 Color distribution module. Given the R_{ab} , its histogram $R_H \in R^{1 \times 1 \times Q}$ is calculated according to the quantization approach described in [38], where Q is the number of color bins for the histogram. Thereafter, three color distribution feature representations with difference spatial sizes and depths are extracted through a set of conv-relu layers and spatial replication operation, as shown in Fig. 4.

3.3.3 Gated fusion module. Without gating mechanism, the colors generated from semantic color assignment module or color distribution module are directly transferred back to the MCN but the conflicts between them tend to cause mis-colorization results. In order to fuse these two types of colors, a confidence map is needed,

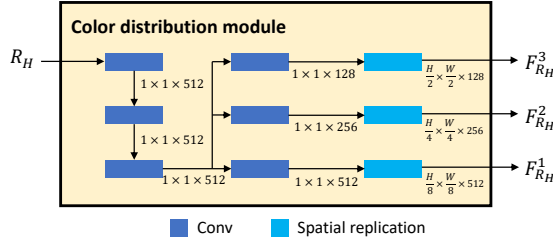


Figure 4: Structure of color distribution module.

which indicates the importance between assigned semantic colors and colors from distribution module. As discussed in color assignment module the correlation matrix reflects the semantic similarity between input image and the reference image. Thus in the proposed framework, the correlation matrix C is used as the input of gated fusion module and the confidence is learned from C by two conv layers, whose kernel sizes are $1 \times \frac{HW}{4}$ and 1×1024 and filter numbers are 1024 and 1 accordingly. Thereafter, the confidence map Ω of different scales can be obtained through reshaping, sampling operations, as illustrated in Fig. 5. Based on the confidence map Ω , the gate operation in this network can be expressed as:

$$M^i = \Omega^i \odot F_S^i + (1 - \Omega^i) \odot F_{RH}^i, \quad (4)$$

where $\odot$ is the element-wise product, and F_S^i and F_{RH}^i are color features extracted from color assignment module and color distribution module.

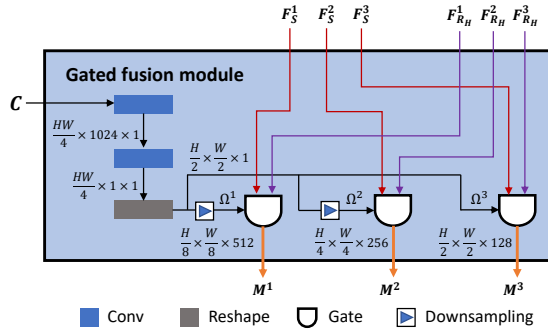


Figure 5: Structure of gated fusion module.

3.4 Losses

The choice of loss functions are very important for network performance. To train the proposed image colorization model, four losses and a regularizer are used in this paper as shown in Fig. 2.

3.4.1 Smooth-L1 loss. Smooth-L1 loss is widely adopted for image colorization to avoid using the average scheme when solving the multi-modal ambiguity problem of colorization [14, 36, 39]. In our work, given the system's output $\hat{T}_{ab}$ and its corresponding ground truth T_{ab} , the smooth-L1 loss $\mathcal{L}_{pixel}(\hat{T}_{ab}, T_{ab})$ can be calculated the same as [39].

3.4.2 GAN's loss. In the training phase, large amount of real color images Y are used with the LS-GAN loss [24] to encourage the proposed image colorization network to produce realistic images,

where the losses of the generator $\mathcal{L}_G$ and discriminator $\mathcal{L}_D$ are defined by:

$$\min \mathcal{L}_G = \frac{1}{2} \mathbb{E}_{\hat{T}_{Lab} \sim \mathbb{P}_T} [(\mathcal{D}(\hat{T}_{Lab}) - 1)^2], \quad (5)$$

and

$$\min \mathcal{L}_D = \frac{1}{2} \mathbb{E}_{\hat{T}_{Lab} \sim \mathbb{P}_T} [\mathcal{D}(\hat{T}_{Lab})^2] + \frac{1}{2} \mathbb{E}_{Y \sim \mathbb{P}_Y} [(\mathcal{D}(Y) - 1)^2], \quad (6)$$

where $\mathbb{P}_T$ and $\mathbb{P}_Y$ are the distributions of output color image $\hat{T}_{Lab}$ and real color images Y , respectively.

In order to fully capture the high-frequency structures from the images, the patch based discriminator [18] is also applied in our framework. The detailed structure of the discriminator can be found in the supplementary.

3.4.3 Color histogram loss. To fully capture the global color distribution R_H of the reference image and convey it to the input grayscale image, which forces the input image to have the same global colors distribution $\hat{T}_H$ as the reference image, the color histogram loss is implemented in the proposed colorization method.

Conventionally, the histogram loss used for image colorization is implemented by representing the output color image by a three-dimensional $(H \times W \times Q)$ tensor whose channels Q denote the bin of histogram by soft-encoding scheme, and histogram loss is computed by multinomial cross entropy [19, 32, 38, 39]. In our work, because the use of GAN's loss, it is infeasible to express output color images using such high-dimensional tensor due to the huge amount of memories usage. Thus, only ab channels are used to represent the produced colorized image in our system, and a differentiable histogram calculation method is required accordingly. To obtain the histogram of an image with its ab channels, the ab color space is normalized to $[-1, 1]$ initially and quantized into bins through a grid whose size is d . And then a differentiable approach is proposed where the i th bin's value of the histogram $\hat{T}_H$ for system's output $\hat{T}_{ab}$ is obtained by:

$$\hat{T}_H(i) = \frac{1}{HW} \sum_{h,w} \frac{1}{d^2} \times \max(0, d - |\hat{T}_{ab}(h, w, a) - m_i|) \times \max(0, d - |\hat{T}_{ab}(h, w, b) - n_i|). \quad (7)$$

where (m_i, n_i) is the grid vertex in ab color space which is corresponding to the quantized center of i th bin. As can be seen from Eq. 7, mapping the image from ab color space to the histogram can be accomplished using the operation similar to bilinear interpolation from which the value $\hat{T}_H(i)$ is calculated by weighted sum of vertex (m_i, n_i) 's neighboring pixels in ab color space. It should be noted this operation is different from the standard bilinear interpolation, which is usually used for calculating the values at random position on the regular grid by its neighboring vertices' values.

Based on the obtained histogram of images, the histogram loss can be defined as the symmetric χ^2 distance[26]:

$$\mathcal{L}_{hist}(\hat{T}_H, R_H) = 2 \times \sum_i \frac{(\hat{T}_H(i) - R_H(i))^2}{\hat{T}_H(i) + R_H(i)}, \quad (8)$$

where Q is the number of color bins for the histogram. In our work, $d = 0.1$ is applied which causes the $Q = 441$.

3.4.4 Classification loss. In the proposed method, classification loss $\mathcal{L}_{class}$ is proposed to improve the capability of the encoder in MCN to extract meaningful semantic features. Given an input grayscale image T_L , we define $\mathcal{L}_{class}$ as the cross-entropy [30]

between the predicted class label $\hat{G}_{T_L}$ and its ground truth label G_{T_L} .

3.4.5 TV regularizer. To encourage spatial smoothness in the output image $\hat{T}_{ab}$, the total variation regularizer [29] is also applied in our work, which is represented as $\mathcal{L}_{tv}(\hat{T}_{ab})$.

3.4.6 Total loss. For the described losses and regularizer, $\mathcal{L}_{pixel}$, $\mathcal{L}_{hist}$, $\mathcal{L}_{\mathcal{G}}$ and $\mathcal{L}_{tv}$ are applied to the three outputs with different scales from the stacked decoders, and $\mathcal{L}_{class}$ is applied to the intermediate output from the encoder in MCN. Based on these losses and regularizer, the total loss for our work is:

$$\mathcal{L}_{total} = \sum_{i=1}^3 \left(\frac{1}{N} \sum_{j=1}^N (\lambda_{pixel} \mathcal{L}_{pixel}^{i,j} + \lambda_{hist} \mathcal{L}_{hist}^{i,j} + \lambda_{tv} \mathcal{L}_{tv}^{i,j}) + \lambda_{\mathcal{G}} \mathcal{L}_{\mathcal{G}}^i \right) + \lambda_{class} \mathcal{L}_{class} \quad (9)$$

where N is the total number of the training samples, j is the index of the image and i indicates output images with different scales from three branches of the stacked MCN, and λ s are the weights for different losses.

4 EXPERIMENTS

4.1 Databases

To train the proposed image colorization model, the images from ImageNet ILSVRC 2012 database [8] are used, which contains more than 1.2 millions of images from 1000 categories. All these images are resized to ensure the short sides of the images are 144 before further processing.

To evaluate the proposed image colorization approach, images with high aesthetics qualities from 8 subcategories in AVA dataset [25]: Animals(ANI), Architecture(AT), City space(CS), Food and Drink(FD), Floral(FL), Land space(LS), Still life(SL) and Portraiture(PT) are used. To build the semantic related subset (subset-1), given an input grayscale image T_L , the image which is from the same category in AVA dataset and is closest to the image T_{Lab} in VGG's feature space is selected as the corresponding reference image R_{Lab} . The semantic independent evaluation set (subset-2) is created simply by randomly selecting a pair of input grayscale image T_L and reference image R_{Lab} from separate categories. Overall, the evaluation set contains 5000 semantic related image pairs and 5000 semantic independent image pairs. All images in this dataset are cropped into size of 128×128 in our experiments.

4.2 Training setup

In this paper, we set $\lambda_{pixel} = 1000$, $\lambda_{hist} = 1$, $\lambda_{class} = 1$, $\lambda_{\mathcal{G}} = 0.1$, $\lambda_{tv} = 10$ for all the experiments. And adam algorithm is used to optimize the network, where the learning rate is 0.0002 and the learning rate decay strategy is not used.

The training of the proposed system is carried out in two stages. In the first stage, only the MCN is trained, where the histogram loss is ignored. We train the MCN using a total of 1.2M images from ImageNet with 10 epochs and batch size of 48, where each training sample provides an input grayscale image T_L along with its corresponding ground truth T_{ab} .

In the second training stage, the entire network with all losses is trained by 10 epochs whose batch size is 12. In order to force the network to lean towards the semantic color when the input and

reference image are semantic related, the corresponding image pairs are constructed, where in each training pair the target image T_{Lab} is randomly cropped from the color image in ImageNet set, and the reference image is produced from the same color image by randomly flipping, rotation, resizing initially and cropping afterward. In this way, the reference image R_{Lab} shares the same content with the input image T_L semantically. Based on T_{Lab} and R_{Lab} , the input T_L , R_{ab} and the ground truth T_{ab} , T_H can be obtained. But the input R_H comes from the image randomly from other category instead. To encourage the network to be biased in favor of color distribution features when similar contents between input and reference images are not available, the semantic independent training set is formed from which we select the target image T_{Lab} from an image in one category of ImageNet, but the reference image R_{Lab} is selected from different categories in ImageNet set. Based on images T_{Lab} and R_{Lab} of each collected pair, the inputs T_L , R_{ab} and the ground truth T_{ab} , T_H can be obtained. In this case, R_H is equal to T_H . For each epoch, the 400K training pairs are randomly generated, where the semantic related pairs and independent pairs are evenly produced. And the sizes of all training images are cropped into 128×128 .

4.3 Comparison with the state-of-the arts

To evaluate the performance of the proposed colorization system Gray2ColorNet, the results of our method are compared to the other state-of-the-art colorization approaches, including the automatic colorization approach [19], color histogram based approach [32, 39] and exemplar based approach [14]. All comparison results are obtained by running the publicly available codes and models provided by the authors.

A qualitative comparison for selected representative cases is shown in Fig. 1. In this figure, one can observe that when semantic related references are provided, Gray2ColorNet along with [14] can effectively transfer color from corresponding objects of reference images to the grayscale images, as shown of colors on lips and faces of the lady in the first row. Whilst the approaches [32, 39] are failed to capture these details but only map overall tones of reference images to the lady's face, as can be seen from the corresponding columns in Fig. 1. On the contrary, when input and reference images are semantically independent, where the correspondence objects between these two images are not available, the color histogram based methods [32, 39] can transfer the tones of reference image to the target image to produce plausible results. However, [14] tends to use prior color knowledge from the database but ignores the colors from the reference image to generate the colorization results, which is similar to the automatic colorization approach [19] in the third column. By using Gray2ColorNet, the tones of reference images can still be transferred to the grayscale images successfully to produce pleasant final outputs. In the second row of Fig. 1, it can be found that the proposed Gray2ColorNet is superior to the semantic color assignment based method [14], when the input image and reference image are semantic related. From the fourth row of this figure, one can see that when the input image and reference image are semantic independent, Gray2ColorNet achieves better colorization result than the other color histogram based approach [32, 39]. These experimental results show the proposed method's robustness for

different types of references. More comparison results can be found in our supplemental material.

In order to quantitatively evaluate the proposed colorization algorithm, the scores of histogram intersection similarity (HIS) [18], which measure the overall similarity of color histogram between reference images and output images produced by different colorization approaches on the evaluation set, are listed in table 1. From this table, it can be seen that the color histogram based methods [32, 39] have better HIS scores than semantic color based approach [14] on semantic independent subset-2, but their HIS scores are lower on semantic related subset-1. But by using the proposed Gray2ColorNet along with the color histogram loss, the colorization results obtain the tones closer to the overall color distribution of reference images than other methods, which achieves the best HIS scores for both subsets. This result shows the superior performance of the proposed colorization algorithm w.r.t. HIS scores.

In addition to qualitative and quantitative comparisons, perceptual metric is also used to assess the attractiveness of our results by human observers. Our method is compared with the-state-of-the-art approaches by user rating based on the quality of colorized pictures.

In this experiment, 50 pairs of evaluation samples from the test dataset are randomly selected, where the semantic related pairs and independent pairs are evenly distributed. Then, the proposed method along with four benchmark methods is used to colorize the input grayscale images based on the provided references. Eventually, total of 250 colorized images from the 50 pairs are obtained. To assess the perceptual quality of the colorized images, all color images generated by different approaches are demonstrated to 22 participants anonymously and randomly. The users are required to score each colorized image given the input and reference image individually, where the integer score ranges from 1 to 5 to indicate the perceptual quality from low to high. Therefore, the final quality score for each colorized image is computed by averaging the score from all participants.

In table 2, the average quality scores for each colorization method are listed. From these numbers, it can be seen that the automatic colorization [19] reaches the lowest perceptual quality score 2.18 due to it ignores the users preference. And we can observe from the table that the proposed method achieve the highest score 3.88, which shows that the proposed Gray2ColorNet can produce color images more pleasing with similar or irrelevant reference images, compared to other exemplar based approaches.

4.4 Network analysis

In order to analyze the effects of color assignment module, color distribution module and gate mechanism in GCFN, three experiments are implemented.

4.4.1 Color assignment module. To evaluate the effectiveness of the color assignment module, we train a colorization system with MCN and color assignment module only through semantic related training pairs. As shown in table 1, the HIS score of this system is 0.67 on the semantic related subset-1, which is 4% higher than the semantic based colorization method in [14]. A sample output from our system is shown in the first row of Fig. 6, which is also compared

with output from [14]. As can be seen from this figure, the curved bill of ibis of input image is colored as red successfully based on the assignment with the reference image, which shows that the proposed color assignment module provides more reliable semantic assignment and color transfer than [14]. This is benefited from the joint learning of semantic correspondence and image colorization. One can see more qualitative comparisons in supplemental material.

From table 1, we can also observe that the colorization performance of color assignment module only system on the semantic independent subset-2 is around 3% lower than the proposed full colorization system, which indicates its lack of capability to transfer colors from reference image to the input image if they are not semantically related.

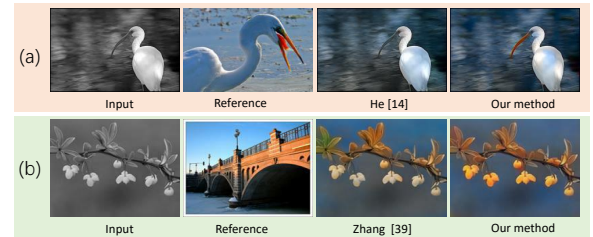


Figure 6: Comparisons for colorization results for (a) color assignment module only and (b) distribution module only.

4.4.2 Color distribution module. To analyze the performance of color distribution module, the network of MCN with color distribution module only is trained by using the semantic independent training pairs. From table 1, it can be seen HIS score of 0.65 is obtained by using color distribution module only on the semantic independent subset-2, which is higher than [39], where the color histogram loss is also applied for colorization. In the second row of Fig. 6, the colorization result with this system is illustrated and compared with [39], from which we can see the proposed color distribution module alone can provide a tone closer to the reference image than [39]. These results show the effectiveness of the proposed color distribution module and the color histogram loss, which can effectively improve the colorization performance. However, due to the lack of semantic information, it can also be seen the system with color distribution module only achieves 0.65 on semantic related subset-1, which is 4% lower than the full colorization system. Interested reader can refer more qualitative comparisons in attached supplemental material.

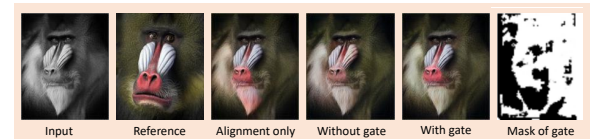


Figure 7: The colorization results with or without gate.

4.4.3 Gate operation. As aforementioned, without gated based fusion mechanism, all the color features from the color assignment module and color distribution module are directly added to the

Table 1: HIS scores of colorization results for different approaches on evaluation datasets.

Eval set	Larsson [19]	Zhang [39]	Xiao [32]	He [14]	color assignment module only	color distribution module only	w/o gated fusion module	w/o classification loss	w/o color histogram loss	full
Subset-1	0.39	0.62	0.59	0.63	0.67	0.65	0.66	0.66	0.62	0.69
Subset-2	0.32	0.63	0.57	0.55	0.63	0.65	0.63	0.65	0.61	0.66
All set	0.36	0.63	0.58	0.59	0.65	0.65	0.65	0.66	0.61	0.67

Table 2: Quality scores of colorization results provided by users for different approaches.

Eval set	Larsson [19]	Zhang [39]	Xiao [32]	He [14]	ours
Semantic related pairs	1.89	2.75	2.47	3.90	4.29
Semantic independent pairs	2.35	3.42	3.08	2.62	3.56
All pairs	2.18	3.1	2.81	3.29	3.88

MCN. In the third sample of Fig. 7, we can observe the semantic correspondence can be established between input and reference images for mandrill’s face, such as the nose and eyes of mandrill. But when the gating mechanism is not available, the colors are ambiguous due to the disturbing of the color information provided by the color distribution module, which causes the colors in the final output image are not accurate, as shown in the fourth sample in Fig. 7.

When the gating mechanism is applied, the proposed colorization system is capable of determining the regions which are semantic related and masking them out, as shown in the last column of Fig. 7, where white areas represent pixels semantically related between input and reference image. By using this mask, the MCN can transfer the semantic colors from reference image to the input image if the pixels are semantic related (such as the nose and face of mandrill), or apply the colors based on the tone of reference image if the pixels are semantic independent between two images (such as the beard of mandrill). The sample output in the fifth column of Fig. 7 demonstrates the correct colorization result, which shows the effectiveness of the proposed gating mechanism.

In table 1, it can also be seen that the overall HIS scores are degraded by 3% on average when the gating mechanism is not available for the colorization system, which shows its effectiveness for selecting and transferring proper colors from reference image to the input image.

These three experiments show that color alignment module and distribution module in the proposed framework can capture the color characteristics of reference image from the semantic and color distribution perspectives. And by using the gate operation, the proposed colorization method can balance the color influence from these two aspects to obtain better colorization performance.

4.5 Analysis of losses

To analyze the effects of classification loss and color histogram loss, two loss functions are ablated separately and their colorization performances are evaluated.

From table 1 we can see when classification loss is removed, the HIS score of the proposed system performance is reduced by 3% on subset-1. In order to analysis the drop of HIS score, the correlation matrix C between input and reference images is applied on the reference image R_{Lab} similar to Eq. 2, by attending to the raw image instead of the feature of the image. As can be seen in

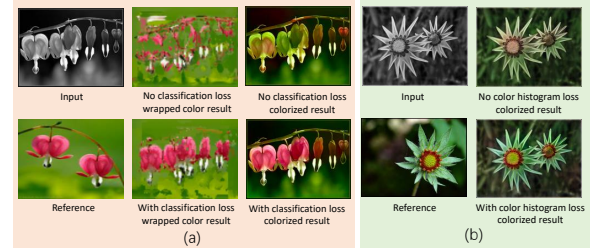


Figure 8: The colorization results with different losses.

the wrapped color results in Fig. 8 (a), the correlation matrix C from the system with classification loss can effectively highlight the corresponding areas between input image and reference image, but the messy wrapped result from the system without classification loss shows that correlation matrix C cannot find the semantic correlation correctly. Based on the obtained C , we can easily see the different colorization performances for the final outputs in the same figure. This observation proves that the classification loss can improve the accuracy of semantic correspondence calculation by enhancing the semantic representation capability of the features output by the encoder of MCN.

In the ablation experiment for color histogram loss, the sample colorized images are demonstrated in Fig. 8 (b), from which one can see that by using color histogram loss, the obtained color image is more vivid than the one without color histogram and its tone is closer to the reference image. This phenomenon is also supported by the histogram intersection similarity (HIS) scores [18] in table 1, which measure the overall similarity of color histogram between input and reference images. From this table, we can see that when color histogram loss is removed, there is a significant decrease in the average HIS score on all set dropping from 0.67 to 0.61. This result demonstrates the proposed histogram loss can significantly improve the color transfer from the reference image to the grayscale image.

5 CONCLUSIONS

In this work, an unified colorization network is proposed where an attention based gating mechanism is applied. Based on the proposed colorization network, the semantic colors and tones of the reference image can be successfully transferred to the input image for both semantic related or independent image pairs. Our method achieves plausible results compared to the state-of-the-art approaches. Need to note that in current stage, our focus is on color transferring effectiveness on low resolution images. Our future work will be applying the method on high resolution images emphasising both effectiveness and efficiency of colorizations.

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